

Conic Sections Formulas

Equations of Parabola:

- 1) Parabola opens up: $(x - h)^2 = 4p(y - k)$ $\vee$
vertex = (h, k) ; focus = $(h, k + p)$; directrix: $y = k - p$
- 2) Parabola opens down: $(x - h)^2 = -4p(y - k)$ $\wedge$
vertex = (h, k) ; focus = $(h, k - p)$; directrix: $y = k + p$
- 3) Parabola opens right: $(y - k)^2 = 4p(x - h)$ $\prec$
vertex = (h, k) ; focus = $(h + p, k)$; directrix: $x = h - p$
- 4) Parabola opens left: $(y - k)^2 = -4p(x - h)$ $\succ$
vertex = (h, k) ; focus = $(h - p, k)$; directrix: $x = h + p$

Ellipse Elongated Horizontally



Equation: $\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1$; $a > b$

$$a^2 = b^2 + c^2$$

Center of Ellipse: (h, k)

Vertices: $(h \pm a, k)$

Foci: $(h \pm c, k)$

Ellipse Elongated Vertically



Equation: $\frac{(x-h)^2}{b^2} + \frac{(y-k)^2}{a^2} = 1$; $a > b$

$$a^2 = b^2 + c^2$$

Center of Ellipse: (h, k)

Vertices: $(h, k \pm a)$

Foci: $(h, k \pm c)$

Hyperbola with Branches Opening Left and Right



Equation of Hyperbola: $\frac{(x-h)^2}{a^2} - \frac{(y-k)^2}{b^2} = 1$

$$c^2 = a^2 + b^2$$

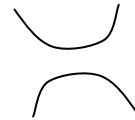
Center of Hyperbola: (h, k)

Vertices: $(h \pm a, k)$

Foci: $(h \pm c, k)$

Equations of asymptotes: $y = \pm \frac{b}{a}(x-h) + k$

Hyperbola with Branches Opening Up and Down



Equation of Hyperbola: $\frac{(y-k)^2}{a^2} - \frac{(x-h)^2}{b^2} = 1$

$$c^2 = a^2 + b^2$$

Center of Hyperbola: (h, k)

Vertices: $(h, k \pm a)$

Foci: $(h, k \pm c)$

Equations of asymptotes: $y = \pm \frac{a}{b}(x-h) + k$